

2.2 Understanding Yield Spreads

1. Following are U.S. Treasury benchmarks available on December 31, 2007:

US/T 3.125 11/30/2009 3.133
 US/T 3.375 11/30/2012 3.507
 US/T 4.25 11/15/2017 4.096
 US/T 4.75 02/15/2037 4.518

On the same day, the following trades were executed:

Issuer	Issue	Yield (%)
Time Warner Cable Inc.	TWC 6.55 05/01/2037	6.373
McCormick & Co. Inc.	MKC 5.75 12/15/2017	5.685
Goldman Sachs Group Inc.	GS 5.45 11/01/2012	4.773

Based on the above, complete the following table:

Issue	Yield (%)	Treasury Benchmark	Benchmark Spread (bps)	Relative Yield Spread	Yield Ratio
TWC 6.55 05/01/2037	6.373				
MKC 5.75 12/15/2017	5.685				
GS 5.45 11/01/2012	4.773				

To finish the above table we first put in the Treasury benchmarks as given in the problem. For 05/01/2037, it is **4.518%**; for 12/15/2017, it is **4.096%**; and, for 11/01/2012, it is **3.507%**.

We can now compute the benchmark spread (bps) given as:

$$\text{benchmark spread} = \text{yield on issuer} - \text{yield on benchmark bond}$$

Inserting our values give the following benchmark spreads:

$$\text{benchmark spread (05/01/2037)} = 6.373\% - 4.518\% = \mathbf{1.855\%}$$

$$\text{benchmark spread (12/15/2017)} = 5.685\% - 4.096\% = \mathbf{1.589\%}$$

$$\text{benchmark spread (11/01/2012)} = 4.773\% - 3.507\% = \mathbf{1.266\%}$$

We can now compute the relative yield spread given as:

$$\text{relative yield spread} = (\text{yield on issuer} - \text{yield on benchmark}) / \text{yield on benchmark}$$

Inserting our values give the following relative yield spreads:

$$\text{relative yield spread (05/01/2037)} = (6.373\% - 4.518\%) / 4.518\% = 41.085\% \text{ or about } \mathbf{41\%}$$

$$\text{relative yield spread (12/15/2017)} = (5.685\% - 4.096\%) / 4.096\% = 38.794\% \text{ or about } \mathbf{39\%}$$

$$\text{relative yield spread (11/01/2012)} = (4.773\% - 3.507\%) / 3.507\% = 36.099\% \text{ or about } \mathbf{36\%}$$

We can now compute the yield ratio given as:

$$\text{yield ratio} = \text{yield on issuer} / \text{yield on benchmark}$$

Inserting our values give the following relative yield spreads:

$$\text{yield ratio (05/01/2037)} = 6.373\% / 4.518\% = 1.41058 \text{ (or about } \mathbf{1.41})$$

$$\text{yield ratio (12/15/2017)} = 5.685\% / 4.096\% = 1.38794 \text{ (or about } \mathbf{1.39})$$

$$\text{yield ratio (11/01/2012)} = 4.773\% / 3.507\% = 1.36099 \text{ (or about } \mathbf{1.36})$$

Below we fill in the missing spaces in bold-face print. We have:

Issue	Yield (%)	Treasury Benchmark	Benchmark Spread (bps)	Relative Yield Spread	Yield Ratio
TWC 6.55 05/01/2037	6.373	4.518	1.855	41%	1.41
MKC 5.75 12/15/2017	5.685	4.096	1.589	39%	1.39
GS 5.45 11/01/2012	4.773	3.507	1.266	36%	1.36

2. The yield spread between two corporate bond issues reflects more than just differences in their credit risk. What other factors would the spread reflect?

In addition to the perceived riskiness of the issuer, other factors that affect the spread include the type of issuer, the term or maturity of the issue, embedded options, taxability of interest received by investors, and expected liquidity of the issue.

3. In the May 29, 1992, *Weekly Market Update* published by Goldman, Sachs & Co., the following information was reported in an exhibit for high-grade, tax-exempt securities as of the close of business Thursday, May 28, 1992:

Maturity (years)	Yield (%)	Yield (%) as a Percentage of Treasury Yield
1	3.20	76.5
3	4.65	80.4
5	5.10	76.4
10	5.80	78.7
30	6.50	82.5

Answer the below questions.

(a) What is meant by a tax-exempt security?

A tax-exempt security is a security in which the investor is exempt from paying certain taxes. For example, purchasers of municipal bond issues are exempt from paying federal taxes. The tax-exempt aspect can affect the spread. For example, part of the spread between Treasury securities and taxable non-Treasury securities of the same maturity reflects the value of the exemption from state and local taxes.

(b) Why is the yield on a tax-exempt security less than the yield on a Treasury security of the same maturity?

The yield on a tax-exempt security is less because investors are excused from paying certain taxes (e.g., federal, state, or local). Thus, they can require a lower rate of return.

(c) What is the equivalent taxable yield?

The equivalent taxable yield is the yield that must be offered on a taxable bond issue to give the same after-tax yield as a tax-exempt issue. In equation form, we have:

$$\text{equivalent taxable yield} = \frac{\text{tax - exempt}}{(1 - \text{marginal tax rate})}.$$

3. What is the after-tax yield for an investor in the 40% tax bracket if the taxable yield is 5%?

The after-tax yield is $0.05 \times (1 - 0.40) = 0.03 = 3\%$

4. What is the taxable-equivalent yield for an investor in the 39% tax bracket if the tax-exempt yield on an investment is 3.1%?

The taxable-equivalent yield is $0.031/(1 - 0.39) = 0.0508 = 5.08\%$

5. Answer the below questions.

(a) What is a yield curve?

The yield curve is the graphical depiction of the relationship between the yield on bonds of the same credit quality but different maturities. The yield curve is usually constructed from observations of prices and yields in the Treasury market.

(b) Why is the Treasury yield curve the one that is most closely watched by market participants?

The Treasury yield curve is the yield curve most closely watched by market participants for several reasons. First, Treasury securities are free from default risk and so differences in credit worthiness do not affect yields making these instruments directly comparable. Second, the Treasury market is the largest and most active bond market offering the fewest problems in terms of illiquidity and infrequent trading.

6. What are the limitations of using Treasury strips to construct the theoretical spot rate curve?

There are three problems with using the observed rates on strips. First, the liquidity of the strips market is not as great as that of the Treasury coupon market. Thus, the observed rates on strips reflect a premium for liquidity. Second, the tax treatment of strips is different from that of Treasury coupon securities. Specifically, the accrued interest on strips is taxed even though no cash is received by the investor. Finally, there are maturity sectors in which non-U.S. investors find it advantageous to trade off yield for tax advantages associated with a strip.

7. What is the common hypothesis about the behavior of short-term forward rates shared by the various forms of the expectations theory?

There are several forms of the expectations theory: pure expectations theory, liquidity theory, and preferred habitat theory. Expectations theories share a hypothesis about the behavior of short-term forward rates are related to expected future short-term rates. These theories also assume that the forward rates in current long-term bonds are closely related to the market's expectations about future short-term rates. These three theories differ, however, as to whether other factors also affect forward rates, and how. The pure expectations theory postulates that no systematic factors other than expected future short-term rates affect forward rates; the liquidity theory and the preferred habitat theory assert that there are other factors. Accordingly, the last two forms of the expecta

8. Give three interpretations of the pure expectations theory.

There are several interpretations of the pure expectations theory that have been put forth by economists. These interpretations are not exact equivalents nor are they consistent with each other, in large part because they offer different treatments of price risk and reinvestment associated with realizing a return.

The broadest interpretation of the pure expectations theory suggests that investors expect the return for any investment horizon to be the same, regardless of the maturity strategy selected. For example, consider an investor who has a five-year investment horizon. According to this theory, it makes no difference if a five-year, 12-year, or 30-year bond is purchased and held for five years because the investor expects the return from all three bonds to be the same over five years. A major criticism of this very broad interpretation of the theory is that, because of price risk associated with investing in bonds with a maturity greater than the investment horizon, the expected returns from these three very different bond investments could differ in significant ways.

A second interpretation, referred to as the local expectations theory, a form of pure

expectations theory, suggests that the returns on bonds of different maturities will be the same over a short-term investment horizon. For example, if an investor has a six-month investment horizon, buying a 5-year, 10-year, or 20-year bond will produce the same six-month return. It has been demonstrated that the local expectations formulation, which is narrow in scope, is the only one of the interpretations of the pure expectations theory that can be sustained in equilibrium.

A third interpretation of the pure expectations theory suggests that the return that an investor will realize by rolling over short-term bonds to some investment horizon will be the same as holding a zero-coupon bond with a maturity that is the same as that investment horizon. (Because a zero-coupon bond has no reinvestment risk, future interest rates over the investment horizon do not affect the return.) This variant is called the return-to-maturity expectations interpretation. For example, let's assume that an investor has a five-year investment horizon. By buying a five-year zero-coupon bond and holding it to maturity, the investor's return is the difference between the maturity value and the price of the bond, all divided by the price of the bond. According to return-to-maturity expectations, the same return will be realized by buying a six-month instrument and rolling it over for five years. Most people have grave problems with the validity of this theory simply due to the unknown nature of future interest rates when assets are rolled over.

9. What is meant by the swap rate?

The fixed interest rate that is paid by the fixed rate counterparty is called the **swap rate**. Dealers in the swap market quote swap rates for different maturities.

10. How does the liquidity preference theory differ from the pure expectations theory?

The pure expectations theory asserts that the only factor affecting the shape of the yield curve is expectations about future interest rates. The liquidity preference theory asserts that there are two factors that affect the shape of the yield curve: expectations about future interest rate and a yield premium to compensate for interest rate risk.

11. If the swap spread for a 5-year interest rate swap is 120 basis points and the yield on the 5-year Treasury is 4.4%, what is the swap rate?

The swap rate is the sum of the 5-year Treasury yield of 4.4% and the swap spread of 120 basis points. The swap rate is therefore 5.6%.

12. Why is the swap spread an important spread measure?

The swap spread is an important spread measure because it is related to credit spreads and therefore can be used in relative value analysis.